Predict a position and a range

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict a future position and explain the limits of the prediction.

SUCCESS CRITERIA - I can...

1. Use matching units and include the starting position.
2. Report a central prediction and its bounds.
3. Distinguish a measurement range from model limitations.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “initial” and “final” to explain your model or result. Sentence starters are optional.

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

bound: an upper or lower limit in a model

01 Bring back an earlier idea

East is positive. Start at 6 m, go to -4 m, then finish at 3 m.
Sketch the route; calculate distance and displacement. Distinguish both from final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $x_f = x_i + vt$; $\delta x_f = \delta x_i + t\delta v$.
Constant velocity; exact positive t and independent bounded starting position/velocity. δ marks a half-width, not a probability.

Worked reasoning: A cart starts at 1 m and moves right at 2 m/s for 3 s. Central prediction: 1+2×3=7 m. If its speed could be 1 to 3 m/s, the endpoint range is 4 to 10 m.

Your turn: Mark the start, central endpoint and endpoint range on a number line. What assumption applies throughout the three seconds?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Units and a starting position

A train starts at x=5 km and moves at +20 m/s for 2 minutes.
Calculate displacement and final position in km.

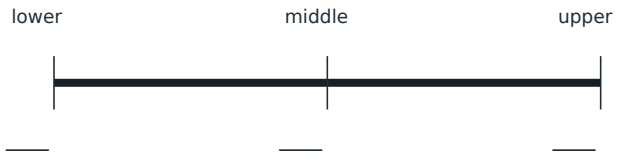
Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

04 Independent bounds

x_0=10±1 m, v=2.0±0.2 m/s, t=5.0 s exact.
Find central position, half-width, endpoints, and full width.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 A negative velocity

x_0=30±2 m, v=-3.0±0.1 m/s, t=10 s exact.
Calculate the full position interval using extreme inputs.

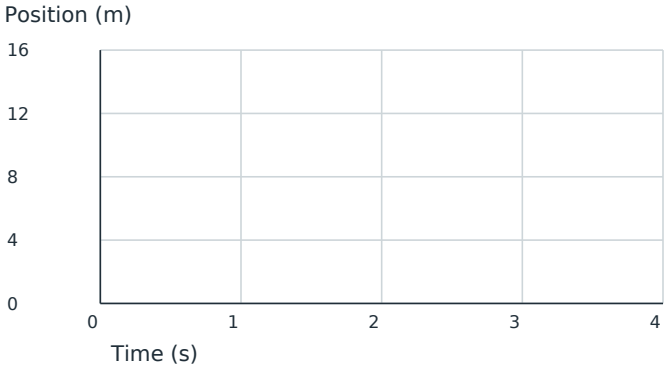
Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Show how the bound grows

For x_0=4±1 m and v=2±0.5 m/s, graph central, lower, and upper x(t) from 0 to 4 s.
Label the endpoint range.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: 1 2 3

07 Input uncertainty or model error?

A precise speed measurement is used to predict an asteroid position with six months of constant velocity.
Explain why tiny measurement bounds do not make the forecast exact.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

08 Choose a method

Velocity is +3 m/s for 5 s, then -2 m/s for 3 s.
Find displacement and distance.
Explain the reference information needed for an angle of 25°.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

09 Exit check

x_0=-5±0.5 m, v=3.0±0.1 m/s, t=10 s exact.
Give central final position, its interval, and the full width.
Name an assumption.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Initial position is 2±0.5 m; constant velocity is 3±0.2 m/s; time is exactly 4 s.
Find the central prediction and worst-case endpoint bounds.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3